

Principles of Microeconomics Notes

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Chapter 1: Ten Principles of Economics

Economists study how individual people make **decisions**, how people **interact** with one another, and the **forces and trends** that affect the economy as a whole.

How People Make Decisions

Principle 1: People face trade-offs

Key Idea

To get one thing we like, we must have to **give up** another — any decision making is **trading off** one goal against another.

Economic examples: Taiwan's **Defense Budget Trade-Off** (Defense can go to defense vs national health insurance)

Key Idea

One main trade-off society faces is: **Efficiency vs Equality**

Economic example: Policies that **redistribute income from the rich to the poor** may increase equality, but decrease efficiency due to the decreased reward for working hard.

Principle 2: The cost of something is what you give up to get it

Key Idea

There is always **opportunity cost**

Economic example: Opportunity cost of **military service** (Conscript pay < civilian job pay, so the opportunity cost is how much one would be “losing” from being conscripted)

Principle 3: Rational people think at the margin

Key Idea

Marginal cost and benefit is what is considered when we make decisions, i.e. **marginal thinking**

Economic examples:

- The **Water–Diamond Paradox** (water is essential, but its plentiful; diamonds are rare, so there's huge marginal benefit) ⇒ actually led to Taiwan running short of water because of TSMC requiring a lot of water supplies, cutting irrigation to farmland (water stopped being plentiful, so its marginal value rose)

- Planes offering a standby seat for far less (marginal cost for one more passenger is tiny if the plane will fly anyway)
- **Netflix** can set the marginal cost of watching one more movie to zero (once the monthly fee is paid, watching another film costs almost nothing extra)

Principle 4: People respond to incentives

Key Idea

Rational people make decisions by weighing costs and benefits, so they must respond when **costs and benefits change**, mainly in the form of **incentives**

Economic examples:

- **Taipei's Trash Policy (Succeeding)**: Households must use the official garbage bag, so **every extra bag now has a real marginal cost**
- **Seat Belt Law (Backfiring)**: Seat belts made accidents less costly by reducing risk of injury, which decreased the benefit of driving safely; due to time and effort taken to drive safely, people drove less safely

How People Interact

Principle 5: Trade can make everyone better off

Key Idea

Trading with others lets each person **specialize in what they do best**, so everyone can buy a greater variety of goods and services at a lower cost than they could achieve alone

Economic example: **Taiwan trading chips for oil, coal, and natural gas**

Principle 6: Markets are usually a good way to organize economic activity

Key Idea

Market economies are more effective than centrally planned economies which lack the information to decide what and how to produce, and for who

Economic example: **Taiwan's high density of convenience stores** (No planning agency decided how many stores each neighborhood needed, each store's owner just responded to the people)

Adam Smith's Invisible Hand

Adam Smith's Invisible Hand steers markets and households towards outcomes that maximize the well-being of society as a whole

Principle 7: Governments can sometimes improve market outcomes

Key Idea

The invisible hand can only work if the government **enforces the rules** a market economy depends on

Without secure property rights, people stop producing:

- A farmer won't grow food if they expect the crop to be stolen before harvest
- A film studio won't produce movies if it cannot stop people from copying them for free

However, the invisible hand can still **fail**, likely from two common sources:

- **Externality**: The effect of one person's actions on a bystander (e.g. pollution)
- **Market power**: A single actor's ability to influence market prices (e.g. monopoly)

Economic example: **Taiwan's National Health Insurance (Successful)**

- A private health insurance market tends to fail (insurers want to avoid the sickest customers; healthy people are tempted to skip coverage until they need it)
- Taiwan's NHI program making **enrollment mandatory for essentially everyone** corrects a market failure (prevents the market from unraveling) and also promotes **equality**

How the Economy as a Whole Works

Principle 8: A country's standard of living depends on its ability to produce goods and services

Key Idea

Raising productivity leads to raising living standards

Economic example: **The Taiwan Miracle** (mostly agricultural economy to an advanced semiconductor production)

Principle 9: Prices rise when the government prints too much money

Key Idea

In almost any case of high or persistent **inflation**, the underlying cause is likely that the **government created money faster** than the **economy producing goods and services**

Principle 10: Society faces a short-run trade-off between inflation and unemployment

Key Idea

Increasing the amount of money in the economy stimulates spending, which encourages companies to hire more and thus lower unemployment. However, this may drive to inflation. $>_<$

Chapter 2: Thinking Like an Economist

The Economist as a Scientist

Key Idea

Economists can rely on **natural experiments** to better understand the economy

Economic examples: War and the Price of Oil (US-Israel war against Iran), COVID-19 (comparing regions that locked down hard vs regions that did not)

The Circular-Flow Diagram

Why do economists use models?

Economists use **simplifying assumptions and models** to focus on the essential features of a complicated economy.

The diagram simplifies the economy down to two decision makers—firms and households—interacting in two markets. It **leaves out government and international trade**.

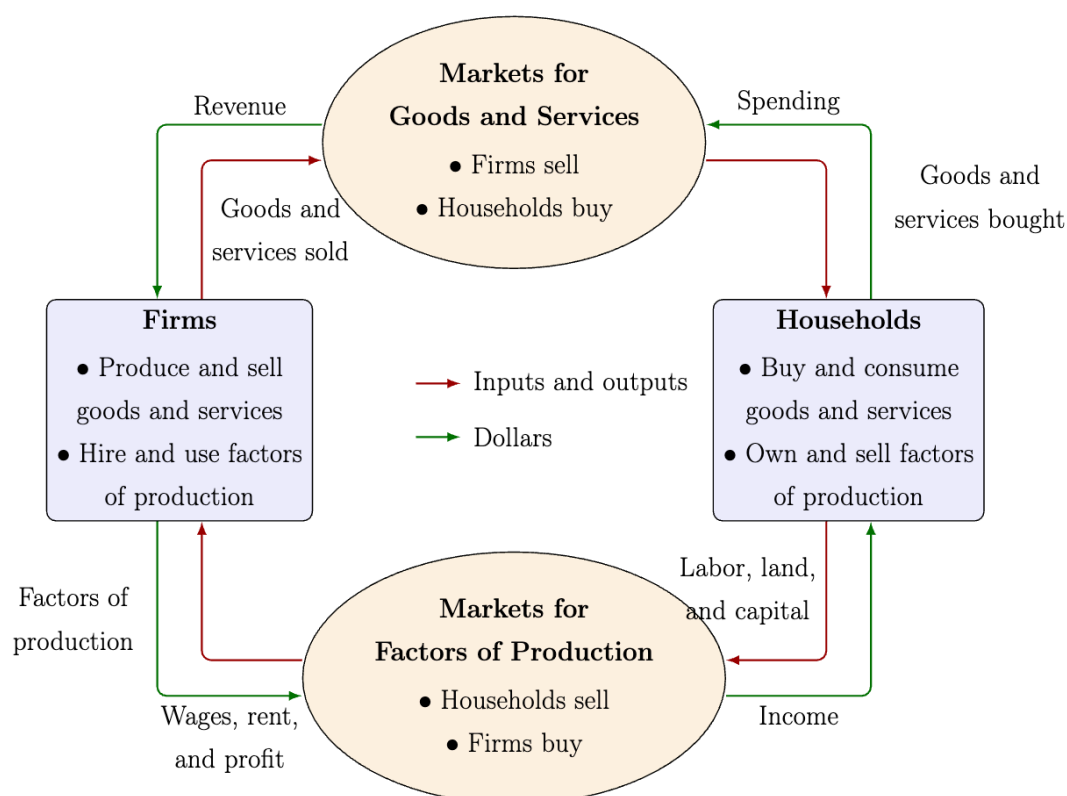


Figure 1: The circular-flow diagram

The Production Possibilities Frontier (PPF)

We can categorize an economy into one of three categories:

- **Efficient**: Sit **on** the frontier
- **Infeasible**: Lie **outside** the frontier — unachievable
- **Inefficient**: Lie **inside** the frontier — some resources are unused

Economic growth shifts the frontier, as shown below:

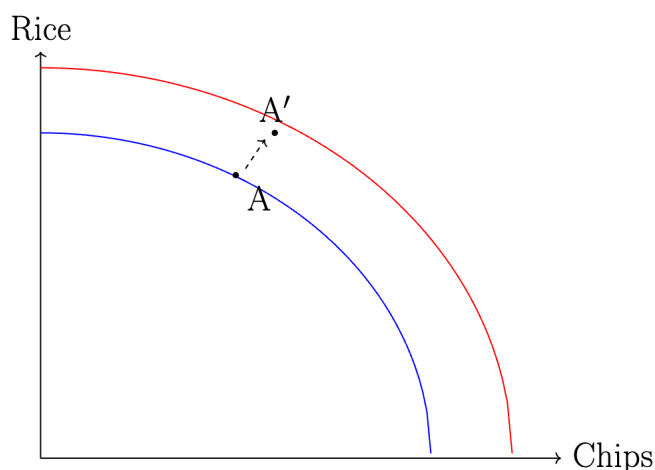


Figure 2: Economic growth shifting the frontier

The Economist as Policy Adviser

Key Idea

When economists explain the effects of a policy, they act as scientists and are objective. When they **recommend** a policy, they act as **policy advisers**, which are subjective.

Positive vs Normative Statements

Key Idea

Economists may **disagree** about **competing positive theories**; they may disagree about **normative statements** since they **hold different values**.

Chapter 3: Interdependence and the Gains from Trade

Main Example

Setting

There are **two countries** (Taiwan and Cambodia), each only produce **two goods** (chip batches and tons of rice), using only one input (labor). We also assume each country switches between two goods at a constant rate, and works for **8 hours a day**. The labor conditions are given below.

	Mins per chip batch	Mins per ton of rice
Taiwan	20	10
Cambodia	60	15

We see, if each country spends 4 hours on each good each day, Taiwan can produce **12 batches of chips** and **24 tons of rice** a day, whereas Cambodia can produce **4 batches of chips** and **16 batches of rice** a day. Although any other point on the PPF is possible. Clearly, **Taiwan has an absolute advantage**.

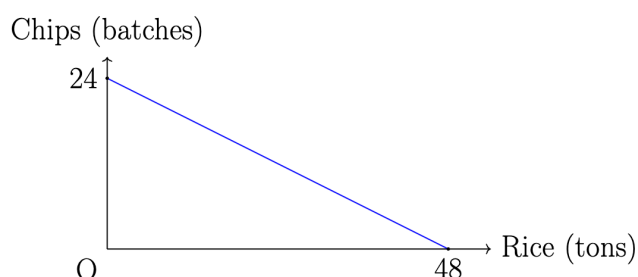


Figure 3: Taiwan's PPF

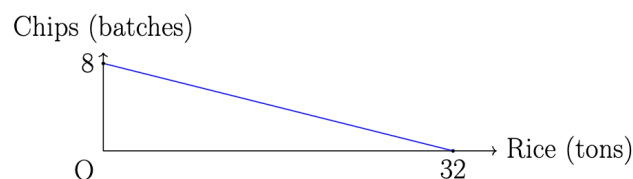


Figure 4: Cambodia's PPF

However, in terms of **opportunity cost**, we see

	Opportunity cost of 1 chip batch	Opportunity cost of 1 ton of rice
Taiwan	2 tons of rice	0.5 chip batches
Cambodia	4 tons of rice	0.25 chip batches

Then, Taiwan has a comparative advantage in chips ($2 < 4$) and Cambodia has a comparative advantage in rice ($0.25 < 0.5$).

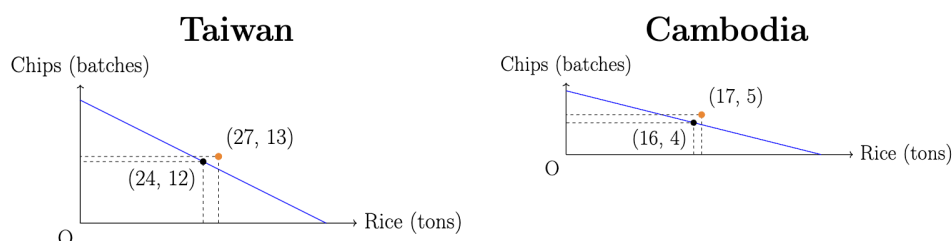
Suppose Taiwan shifts more time towards chips, producing **18 chip batches and 12 tons of rice** a day, and Cambodia specializes completely in rice, producing **32 tons of rice and no chips**.

Suppose Cambodia trades 15 tons of rice to Taiwan in exchange for 5 chip batches (3 tons of rice per chip batch). This lies in **between Taiwan's opportunity cost of a chip batch (2 tons of rice) and Cambodia's (4 tons of rice)**, which is **exactly the range** which both countries are willing to trade.

Notice, after trading, both countries get to be above their PPFs, which is a **gain for both sides**.

Summary

Trade allows for mutual benefit due to **comparative advantages** comparing **opportunity costs**, instead of absolute advantages. Trade lets each country consume **outside its own PPF**, a combination of goods it could have never produced by itself.



- without trade (on the frontier)
- with trade (outside the frontier)

Figure 5: Taiwan and Cambodia's PPF vs after trading

Remark

It is impossible for one country to have a comparative advantage in both goods, since the opportunity cost of one good is the inverse of the opportunity cost of another. A country that is relatively cheap at producing one good is automatically relatively expensive at producing the other.

Other Examples

- The surgeon and her assistant: A surgeon who can type faster than anyone else in her office should still hire an assistant to handle her scheduling and paperwork, since every hour she spends on typing is an hour she is not spending on surgery (i.e. opportunity cost of an hour of typing is far higher)
- Taiwan's Rice Import Quota: It raises the country's total economic pie to import rice, but it **can hurt specific producers** (i.e. domestic rice farmers) $\Rightarrow$ **Sometimes it's important to protect local producers, rather than maximizing purely for consumer benefit**

Shapes of PPFs

PPFs are **straight lines** if a country *switches* between producing its two goods at **a constant rate**.

Real economies oftentimes have PPFs that **bow outward** instead of running straight, where the **opportunity cost rises** the more a good is produced.

Constant opportunity cost Increasing opportunity cost

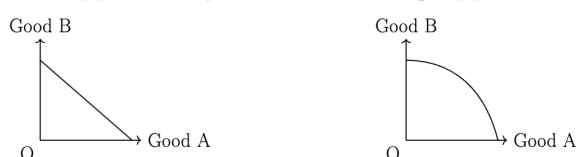


Figure 6: Comparison of different PPF shapes

Chapter 4: Market Forces of Supply and Demand

Demand

The demand curve slopes downward, because other things equal, a lower price means a greater quantity demanded. The demand curve shows how price affects quantity demanded, holding constant everything else that affects buyers.

- A **change in price** moves us along a **fixed demand curve**
- A change in anything else that affects demand, income, the price of related goods, tastes, expectations, or the number of buyers, **shifts the entire curve** to a new position

Shifts in Demand

Any change that raises the quantity demanded at every price shifts the curve to the **right**, which is an **increase in demand**. Any change that lowers the quantity demanded at every price shifts the curve to the **left**, which is a **decrease in demand**.

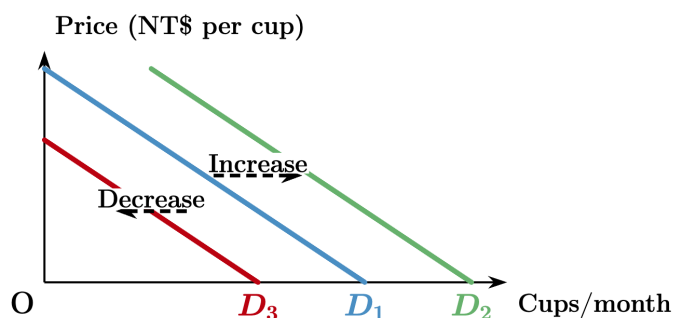


Figure 7: Shifts in the demand curve

What shifts the demand curve?

- **Income**: For a *normal good*, higher income raises demand (e.g. diamonds); for an *inferior good*, higher income lowers demand (e.g. instant ramen)
- **Prices of related goods**: A higher price of a **substitute** (e.g. coffee, for bubble tea) raises demand; a higher price for a **complement** (e.g. tapioca pearls, for bubble tea shops' costs) lowers demand for the other good
- **Tastes**: Anything that makes a good more or less appealing shifts demand in the same direction
- **Expectations**: Expecting **higher prices or higher income in the future** can raise demand today
- **Number of buyers**: More buyers in the market raises demand at every price

Normal and Inferior Goods

Most goods are **normal goods**. For example, if higher income leads a student to buy more bubble tea at each price, then bubble tea is a normal good for that student.

An **inferior good** is the *exception*. As a student's income rises, they may buy instant ramen less, because they may eat proper food more.

Case Study: Two Way to Reduce Smoking

As smoking causes illness, governments often want people to smoke less. This model provides two options:

1. **Shift the demand curve** for cigarettes to the left (e.g. graphic health warnings on packs, bans on tobacco advertising, and public health campaigns $\Rightarrow$ try to reduce the quantity demanded at any price)
2. **Raise the price** (move **along the demand curve**): E.g. Taiwan raised its tobacco tax by NT\$20 per pack, to the extent that it raises the price buyers pay, hence it reduces quantity demanded, while other demand factors are fixed

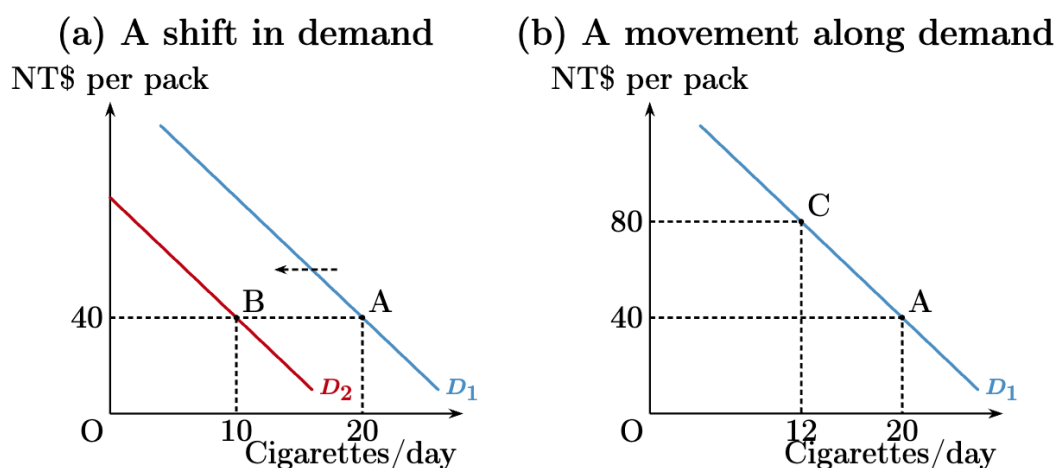


Figure 8: Diagrams representing the shift in the demand curve for the examples above

Supply

The supply curve slopes **upward** because, other things being equal, a higher price means a greater quantity supplied.

Shifts in Supply

Any change that raises the quantity supplied at every price shifts the curve to the **right**, an **increase in supply**. Any change that lowers the quantity supplied at any price shifts the curve to the **left**, a **decrease in supply**.

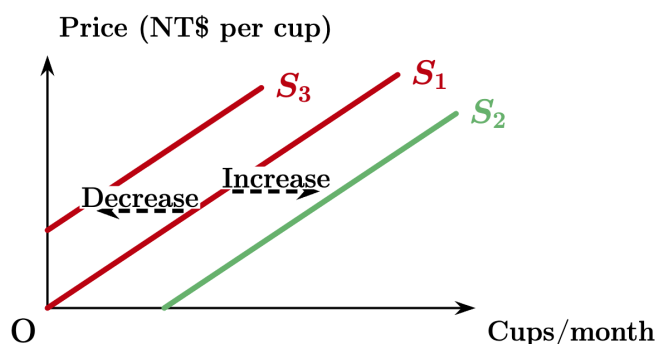


Figure 9: Shifts in the supply curve

What shifts the supply curve?

- **Input prices**: A higher price for an input makes production less profitable and shifts supply to the left
- **Technology**: An improvement that lowers production costs shifts supply to the right
- **Expectations**: Expecting a higher price tomorrow can lower supply today, as sellers of storable goods hold back inventory
- **Number of sellers**: More sellers in the market raises supply at every price, and sellers leaving the market lowers it

Equilibrium

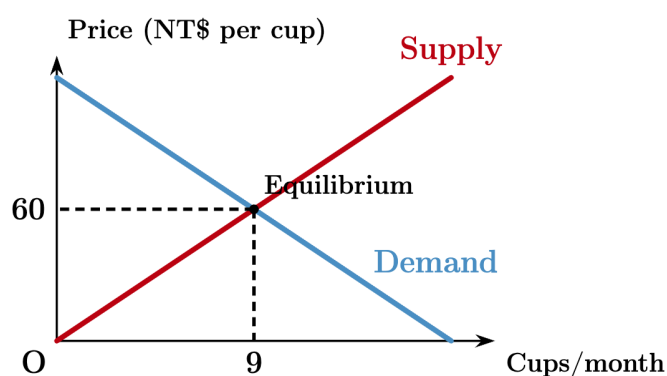


Figure 10: Equilibrium as a graph

Surplus

Key Idea

When there is a surplus of a good, sellers would **lower their price**, which raises quantity demanded and lowers quantity supplied, until the price reaches equilibrium.

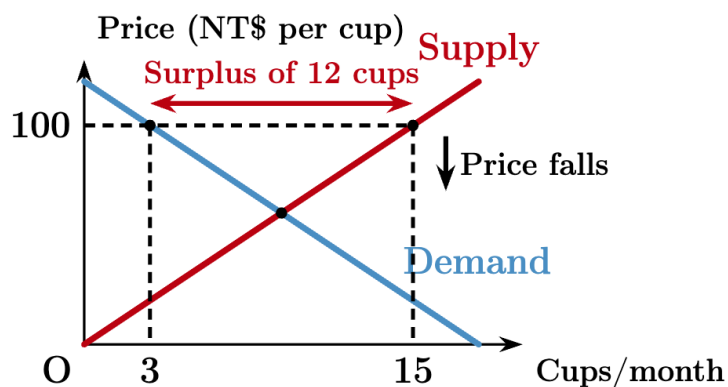


Figure 11: Surplus as a graph

Shortage

Key Idea

When there is a shortage of a good, sellers would **raise their price**, which lowers quantity demanded and increases quantity supplied, until the price reaches equilibrium.

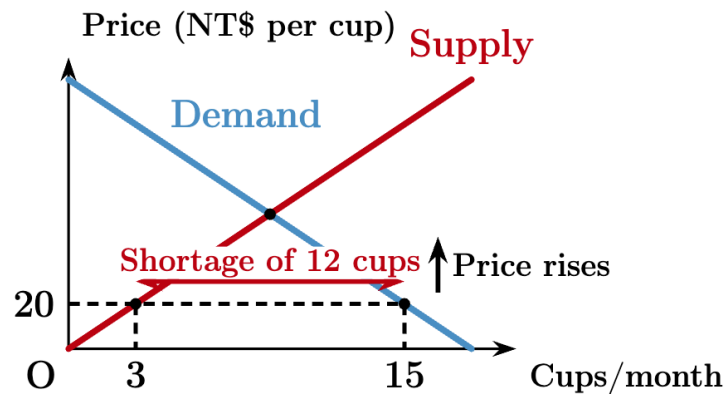


Figure 12: Shortage as a graph

Law of Supply and Demand

Key Idea

At equilibrium, there is no excess demand or supply pushing the price to change, but a new shock can shift the curve and change the equilibrium. **Adjustment takes time**. A binding price control can prevent the price from *clearing* the market, so a shortage or surplus may persist

How Prices Allocate Resources

Example of Price Gouging: Price Gouging and Taiwan's Mask Rationing

Even though the demand for masks increased due to the pandemic, the government didn't want the supply to simply increase and cause the prices to increase. This effect is known as price gouging.

Thus, Taiwan distributed masks, as requested by the government, through pharmacies using National Health Insurance cards. Taiwan also increased production to ease scarcity. This allowed for **higher supply** of masks, which makes the new equilibrium point of maintaining the same price to be possible.

Rationing allowed scarce masks without relying only on willingness to pay.

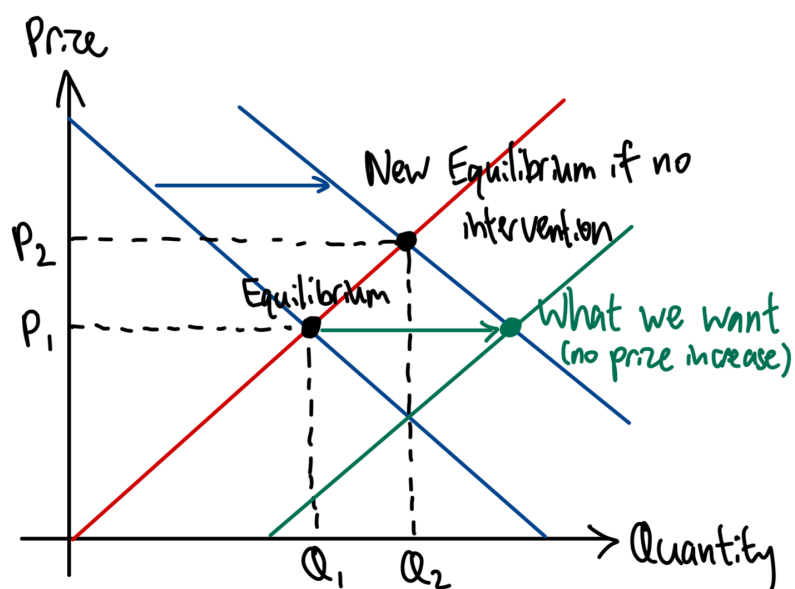


Figure 13: Mask gouging example's supply and demand

Key Idea

Market prices coordinate decisions about **how scarce resources** are used.

Three Steps for Analyzing a Change

When some *event affects a market*, the equilibrium changes, and we get a new price and a new quantity.

1. Decide whether the event shifts the supply curve, demanded curve, or both
2. Decide which direction each affected curve shifts
3. Compare the new equilibrium to the old one using the supply-and-demand diagram

Note that sometimes the shift can be *ambiguous*.

Chapter 5: Elasticity and Its Application

The Price Elasticity of Demand

The **price elasticity of demand** measures how much the **quantity demanded** of a good responds to a **change in its price**. Informally speaking, it measures **how price-sensitive buyers are**.

Factors that Determine the Price Elasticity of Demand

- **Availability of close substitutes**: Goods with close substitutes have more elastic demand, because buyers can easily switch. (E.g. if the price of one brand of instant noodles rises, we can buy another brand)
- **Necessities vs luxuries**: Necessities tend to have less elastic demand than luxuries (E.g. even if the cost of a doctor's visit rises, we still have to go just as often)
- **Definition of the market**: Demand is usually more elastic to a **narrowly defined good**
- **Time horizon**: Demand is usually more elastic over longer periods (E.g. when gasoline gets more expensive, drivers can cut back only a little in the first few months, over years, they can move closer to work)

The Variety of Demand Curves

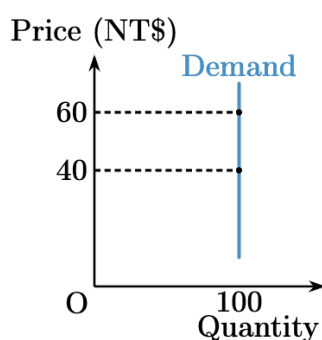
Key Idea

Economists classify demand curves according to their **elasticity**.

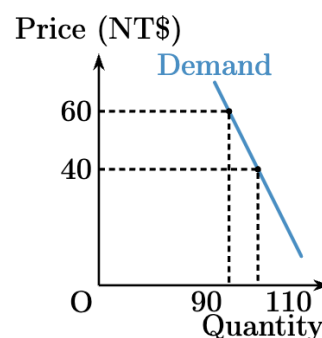
- Demand is **elastic** when the **elasticity > 1** , so quantity moves proportionally **more** than price
- Demand is **inelastic** when **elasticity < 1** , so quantity moves proportionally **less** than price
- Demand is **unit elastic** when **elasticity $= 1$** , so quantity and price change by **equal percentages** in absolute value

Remark: With the same axis units, the **flatter** the demand curve, the **greater the price elasticity of demand**. The **steeper** it is, the **smaller the price elasticity of demand**.

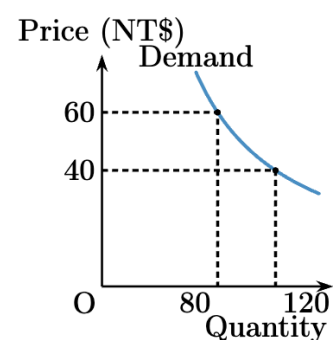
(a) Perfectly inelastic:
elasticity 0



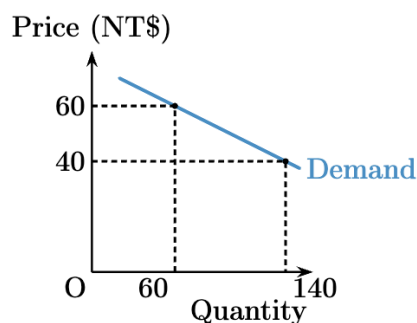
(b) Inelastic: elasticity
below 1



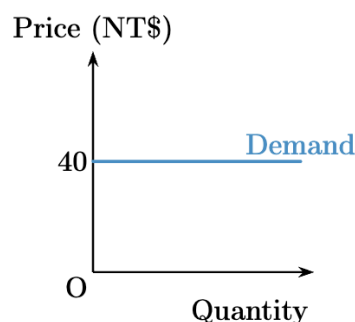
(c) Unit elastic:
elasticity 1



(d) Elastic: elasticity above 1



(e) Perfectly elastic: elasticity infinite



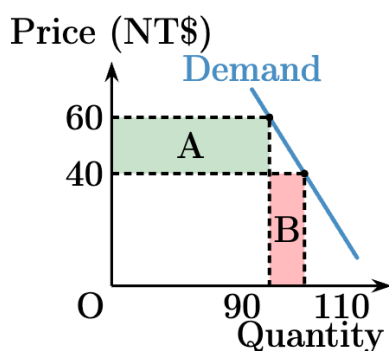
Total Revenue

Key Idea

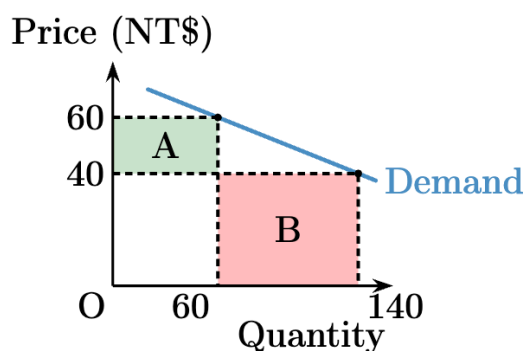
Total revenue is the amount sellers receive from sales: the price received per unit times the quantity sold $P \times Q$. Without a *tax wedge*, this also equals buyers' total spending. Graphically, it's the **area of the box** under the demand curve.

Here, we can see how total revenue (i.e. shaded area) changes when price changes, according to if demand is inelastic or elastic:

(a) Inelastic demand



(b) Elastic demand



The Elasticity–Revenue Rule

Along a fixed demand curve, **elasticity** determines whether the **price effect** or the **quantity effect** on total revenue, $P \times Q$, is **stronger**.

Demand	Price rises	Price falls
Inelastic ($E < 1$)	Revenue rises	Revenue falls
Unit elastic ($E = 1$)	Revenue unchanged	Revenue unchanged
Elastic ($E > 1$)	Revenue falls	Revenue rises

Summary

With inelastic demand, price and revenue move in the same direction. With elastic demand, they move in opposite directions.

Notice that when asked about if the total revenue of a good would increase or fall, the price change alone does not determine the answer—make sure to talk about the elasticity.

Elasticity along a Linear Demand Curve

A straight line has constant slope, but **not constant elasticity**, since elasticity is a ratio of percentage changes.

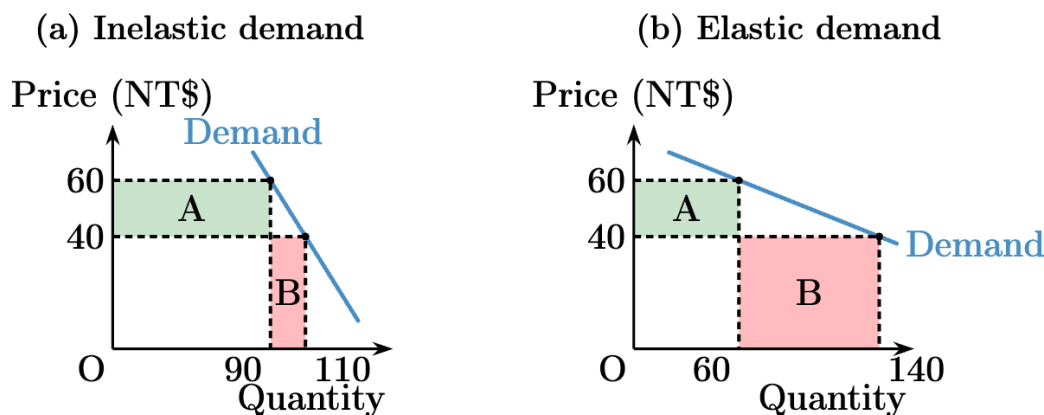
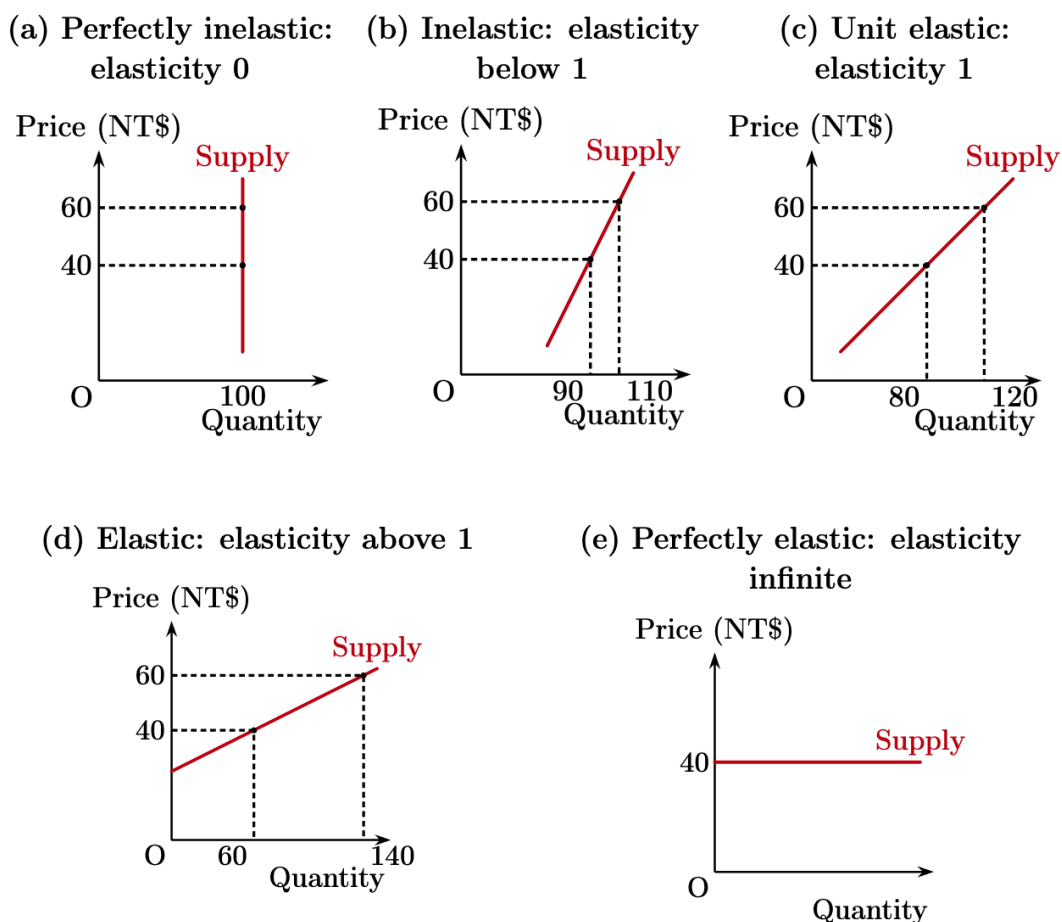


Figure 14: Elasticity along a Linear Demand Curve

The Price Elasticity of Supply

The **price elasticity of supply** measures how much the **quantity supplied** of a good **responds to a change in its price**. Informally, it is **how flexible** sellers are in changing the amount they produce.



Applications

Consider a hypothetical competitive rice market with no imports, exports, or government price support. Suppose a new rice variety lowers production costs without changing the rice's quality. Farmers can supply more rice at every price, so **supply shifts right**. The **equilibrium price falls** and **quantity rises**.

Farmers sell more rice but receive less per kilogram. However, rice may have low demand elasticity, so the **total revenue may decrease**. Although adopting the new variety is individually profitable at the original market price, when all farmers adopt it, the resulting increase in market supply drives the equilibrium price down. The effect on profit still depends on the reduction in production costs.

Key Idea

What is optimal for each individual may not be optimal for the group as a whole.